From Drawing Product to Drawing Process: Combining 2D Sketch Analysis with Video-Based 3D Hand Kinematics for Parkinson's Motor Assessment

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***Abstract*—Parkinson's disease (PD) affects over 10 million people worldwide, with motor symptoms commonly assessed through drawing and handwriting tasks. Current digital assessment requires specialized tablets costing $500–$2000, limiting at-home monitoring. We present a two-part feasibility study combining 2D drawing product analysis with a novel 3D video-based hand kinematic process analysis for PD motor assessment.**

In Part A, using a public dataset of 204 drawings (102 PD patients, 102 healthy controls), we demonstrate that computational sketch features—including image entropy ($p = 0.0003$), stroke density ($p = 0.009$), and crossing count ($p = 0.001$)—significantly distinguish PD from healthy drawings. Six of 16 proposed features achieve statistical significance, and SVM classification reaches 69.7% accuracy on wave drawings.

In Part B, we propose a novel approach using smartphone video-based 3D hand tracking (MediaPipe) to analyze hand kinematics *during* the drawing act. In a proof-of-concept study with 2 healthy participants performing spiral tracing and sentence writing under 5 simulated PD conditions (normal, mild tremor, strong tremor, bradykinesia, micrographia), we find: (1) tremor-related kinematic features increase dramatically under simulated tremor (PD-band power: $+1431\%$, normalized jerk: $+2101\%$, wrist range of motion: $+248\%$); (2) simulated micrographia is detectable from hand kinematics alone, with spectral arc length smoothness showing statistically significant difference ($p = 0.019$, Cohen's $d = 3.2$); and (3) the 3D process approach captures motor information invisible to 2D product analysis.

This combined framework bridges validated sketch product analysis (hardware-dependent) with novel sketch process analysis (requiring only a standard smartphone camera), establishing technical feasibility for accessible PD motor monitoring.

***Index Terms*—Parkinson's disease, hand tracking, sketch analysis, motor assessment, tremor detection, MediaPipe, smartphone sensing, micrographia, SPARC, rehabilitation monitoring**

I. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder, affecting over 10 million people worldwide [1]. The cardinal motor symptoms—resting tremor (4–6 Hz), bradykinesia, rigidity, and postural instability—progressively impair fine motor control. Handwriting and drawing tasks are among the earliest and most sensitive clinical indicators of motor decline [2].

Clinical assessment relies on the Unified Parkinson's Disease Rating Scale (UPDRS) [3], administered during clinic visits every 2–4 months. Between visits, patients lack objective motor monitoring, making medication adjustment reactive. Drawing tasks—particularly the Archimedes spiral test and handwriting analysis for micrographia—are standard clinical tools [4], yet their scoring remains largely subjective or requires specialized tablets costing $500–$2000 [5].

Two fundamentally different approaches exist for computational drawing analysis: (1) *product analysis*—examining the final drawn output, and (2) *process analysis*—examining how the drawing was produced. Product analysis has been extensively studied [5], [6]. Process analysis, which captures richer temporal and kinematic information, has been limited to specialized hardware. Recent advances in markerless hand tracking— particularly Google's MediaPipe framework [7], which provides real-time 3D estimation of 21 hand joint positions from monocular RGB video—now enable process analysis using only a standard smartphone camera.

Research Gap. While 2D drawing product analysis for PD is well-established, no study has investigated: (1) whether video-based 3D hand kinematics during drawing can detect PD motor symptoms, or (2) how product and process analysis complement each other for comprehensive motor assessment.

Contributions. This paper presents:

1) A two-part framework combining 2D drawing product analysis (validated on 204 real PD/healthy drawings) with novel 3D video-based hand kinematic process analysis.
2) A 16-feature motor assessment framework for each modality—Part A features derived from drawing images; Part B features from 3D hand joint trajectories.
3) Demonstration that video-based hand kinematics detect simulated PD symptoms with large effect sizes, and that micrographia is detectable from hand movement alone without analyzing written text ($p = 0.019$, Cohen's $d = 3.2$).
4) Evidence that process analysis captures motor information invisible to product analysis, enabling future accessible, hardware-free PD monitoring.

II. Related Work

A. Drawing and Handwriting Analysis for PD

Drawing tasks have been used for PD motor assessment for decades [4]. The Archimedes spiral test is widely used to quantify tremor severity, with features such as drawing speed, smoothness, and deviation from template correlating with UPDRS motor scores [15]. Micrographia—progressive reduction in handwriting size—affects 50–60% of PD patients and often precedes clinical diagnosis [16].

Digital tablet-based systems have enabled quantitative analysis of pen dynamics including pressure, velocity, and jerk [5], [6]. Drotar et al. [5] demonstrated that kinematic features from digitized handwriting could distinguish PD patients from controls with over 80% accuracy, but these systems require specialized hardware unavailable in home settings.

B. Sketch Recognition and Computational Drawing Analysis

Computational analysis of the drawing process has a rich history in the sketch recognition community. Early work by Rubine [8] established gesture feature extraction from pen trajectories, and Gross [9] examined the ambiguity of hand-drawn marks as a fundamental property of freehand input. Hammond and Davis [11] introduced the LADDER language for formally describing and recognizing hand-drawn sketch primitives, enabling systematic analysis of drawing structure. Paulson and Hammond [12] advanced geometric sketch recognition with accurate primitive recognition and beautification, while Fonseca et al. [10] demonstrated robust sketch understanding from noisy hand-drawn input.

A particularly relevant line of work examines not just the sketch product but the drawing process as a signal of cognitive and motor ability. Hammond et al. [13] proposed metrics for modeling expert sketching ability that capture fluency, temporal dynamics, and process characteristics—features conceptually parallel to our motor health assessment approach. Separately, rehabilitation systems such as Kinohaptics [14] have shown that continuous motor monitoring with automated haptic feedback significantly improves post-surgical recovery, motivating similar approaches for neurodegenerative conditions.

C. Video-Based Hand Analysis

MediaPipe Hands [7] provides real-time 3D estimation of 21 hand landmarks from monocular video. While extensively applied to gesture recognition and sign language [17], its use for clinical motor assessment remains largely unexplored. Yan et al. [18] recently applied MediaPipe to hand tremor quantification. However, no study has combined video-based hand tracking with sketch-process analysis for PD motor symptom detection during functional drawing tasks.

III. Methodology

Our framework consists of two complementary approaches (Fig. 1): Part A analyzes the drawing *product* (what was drawn) using image processing on a public clinical dataset, while Part B analyzes the drawing *process* (how it was drawn) using video-based 3D hand tracking.

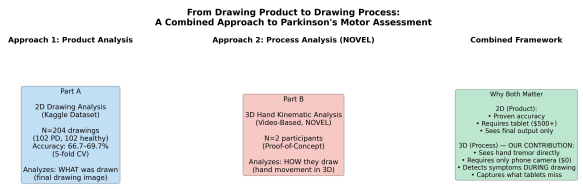


Fig. 1. Combined framework: Part A analyzes the 2D drawing product using image features (validated on real PD patients). Part B analyzes the 3D hand drawing process using video-based kinematics (novel; requiring only a smartphone camera).

A. Part A: 2D Drawing Product Analysis

1) Dataset: We use the publicly available Parkinson's Drawing Dataset [23], containing 204 drawings from 102 PD patients and 102 age-matched healthy controls. Each participant produced both spiral and wave drawings (102 per type). All drawings were scanned at consistent resolution.

2) Feature Extraction: We propose 16 image-based features:

- **Tremor features**: Contour smoothness (curvature variability), tremor intensity (high-frequency spectral power), stroke irregularity (deviation from smooth fit).
- **Kinematic proxy features**: Stroke density, speed proxy (inter-point distance), speed variability.
- **Shape features**: Drawing area, compactness, radial deviation, crossing count, Hu moments (1st and 2nd).
- **Texture features**: Stroke thickness variation, mean thickness, perimeter-area ratio, image entropy.

3) Classification: Three classifiers were evaluated using stratified 5-fold cross-validation: Random Forest (100 trees, max depth 10), Gradient Boosting (100 estimators, max depth 5), and SVM with RBF kernel ($C = 1.0$, $\gamma = \texttt{scale}$). All features were z-score normalized.

B. Part B: 3D Video-Based Hand Kinematic Analysis

1) Participants and Setup: Two healthy participants (both male, ages 23 and 24) provided informal consent for this proof-of-concept study. A consumer smartphone (rear camera, 1080p at 30–60 fps) was mounted 20–25 cm above the drawing surface. Participants drew on white A4 paper under consistent overhead lighting. No specialized hardware was used.

2) Drawing Tasks:

- **Spiral tracing**: Tracing a printed Archimedes spiral template (3 loops, 10 cm diameter).
- **Sentence writing**: Writing "Today is a beautiful day" on blank paper.

3) Simulated Symptom Conditions: To provide grounded kinematic reference ranges prior to clinical data collection, we used established physical simulation approaches [21]:

- **Normal**: No manipulation (baseline).
- **Mild tremor**: Vibrating motor device loosely secured near the drawing wrist to induce low-amplitude oscillation.
- **Strong tremor**: Same device firmly applied to increase oscillation amplitude.

TABLE I
PART A: STATISTICALLY SIGNIFICANT FEATURES (PD VS. HEALTHY)

Feature	Healthy	PD	p	d
Image Entropy	5.159 ± 0.23	5.291 ± 0.29	0.0003***	0.51
Crossing Count	44.6 ± 53.8	93.7 ± 144.3	0.0015**	0.45
Speed Proxy	1.171 ± 0.02	1.177 ± 0.02	0.0062**	0.39
Stroke Density	0.061 ± 0.02	0.070 ± 0.03	0.0090**	0.37
Hu Moment 1	1.942 ± 0.15	2.003 ± 0.19	0.0109*	0.36
Perim.–Area Ratio	50.6 ± 15.1	45.1 ± 20.1	0.0300*	0.31

$*p < 0.05$, $**p < 0.01$, $***p < 0.001$. d: Cohen's d. Values: mean $\pm$ SD.

- **Bradykinesia**: Thick elastic wrap around the wrist to restrict range of motion.
- **Micrographia**: Participant instructed to write with progressively decreasing letter size [16].

Each condition lasted approximately 20–50 seconds per task. Total recordings: 12 videos (2 participants $\times$ 6 conditions).

4) Hand Tracking Pipeline: Videos were processed using MediaPipe Hands [7], extracting 21 three-dimensional hand joint positions per frame. Detection and tracking confidence thresholds were set to 0.5. Mean detection rate across all videos was 63.2% (range: 10.7%–95.6%).

5) Feature Extraction: We propose 16 hand kinematic features organized by PD symptom domain:

- **Bradykinesia**: Mean speed, speed variability, peak speed, normalized jerk [20], spectral arc length smoothness (SPARC) [19].
- **Tremor**: Tremor amplitude (3–12 Hz bandpass RMS), tremor frequency, PD-band power ratio (4–6 Hz).
- **Stability**: Grip stability (thumb-index distance variance), hand steadiness (wrist variability), wrist range of motion, finger coordination (inter-finger velocity correlation).
- **Temporal**: Drawing duration, pause ratio, velocity peaks per second, amplitude decay slope (micrographia index).

Let $\mathbf{p}_t \in \mathbb{R}^3$ denote the index fingertip position at frame t. The SPARC smoothness metric is computed as:

$$\text{SPARC} = -\int_0^{\omega_c} \sqrt{\left(\frac{1}{\omega_c}\right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega}\right)^2}\, d\omega \qquad (1)$$

where $\hat{V}(\omega)$ is the normalized Fourier magnitude spectrum of the velocity profile, and ω_c is the cutoff frequency (10 Hz) [19]. More negative SPARC values indicate smoother (healthier) movement.

IV. RESULTS

A. Part A: 2D Drawing Product Analysis

1) Statistical Analysis: Independent samples t-tests revealed 6 of 16 features significantly differing between PD and healthy groups ($p < 0.05$). Table I presents results for all significant features.

Image entropy was the most discriminative feature ($p = 0.0003$, Cohen's $d = 0.51$), with PD drawings exhibiting higher entropy (more irregular patterns). Crossing count was significantly elevated in PD ($p = 0.0015$), consistent with tremor-induced stroke crossings.

TABLE II
PART A: CLASSIFICATION ACCURACY (%, STRATIFIED 5-FOLD CV)

Task	RF	GB	SVM
Spiral	68.7 ± 7.0	62.8 ± 3.6	66.7 ± 5.7
Wave	68.6 ± 4.2	69.6 ± 5.5	$\mathbf{69.7 \pm 4.3}$
Combined	66.7 ± 6.2	—	—

RF: Random Forest. GB: Gradient Boosting. SVM with RBF kernel.

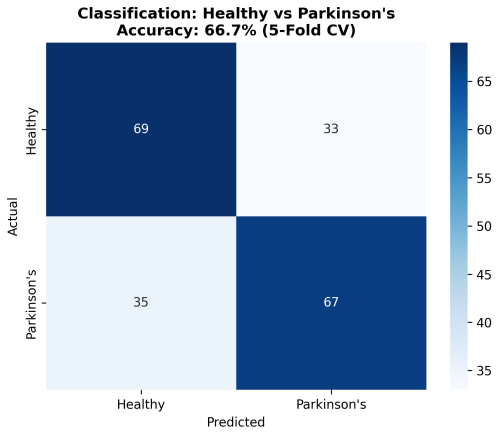


Fig. 2. Part A: Confusion matrix (5-fold CV, combined spiral and wave, Gradient Boosting, 66.7% accuracy).

2) Classification Results: Table II summarizes classification accuracy.

Best single-task performance was achieved by SVM on wave drawings (69.7%). Fig. 2 shows the confusion matrix for the combined classifier.

3) Feature Importance: Fig. 3 shows the Random Forest feature importance ranking. Entropy, stroke density, and speed proxy were the top-3 discriminative features.

B. Part B: 3D Video-Based Hand Kinematic Analysis

1) Hand Tracking Performance: MediaPipe tracked the hand in 63.2% of frames on average (range: 10.7%–95.6%). Higher detection rates correlated with better lighting and reduced motion blur. Videos with detection rates $> 50\%$ (10 of 12) provided sufficient temporal resolution for reliable feature extraction.

2) Tremor Detection: Comparing the normal baseline with simulated mild tremor revealed substantially elevated kinematic features (Fig. 4):

- Normalized jerk: $+2101\%$ increase.
- PD-band power ratio (4–6 Hz): $+1431\%$ increase.
- Peak speed: $+466\%$ increase.
- Wrist range of motion: $+248\%$ increase.
- Speed variability: $+167\%$ increase.
- Tremor amplitude (3–12 Hz RMS): $+140\%$ increase.

3) Micrographia Detection: Comparing normal writing ($n = 4$ recordings) with simulated micrographia ($n = 2$),

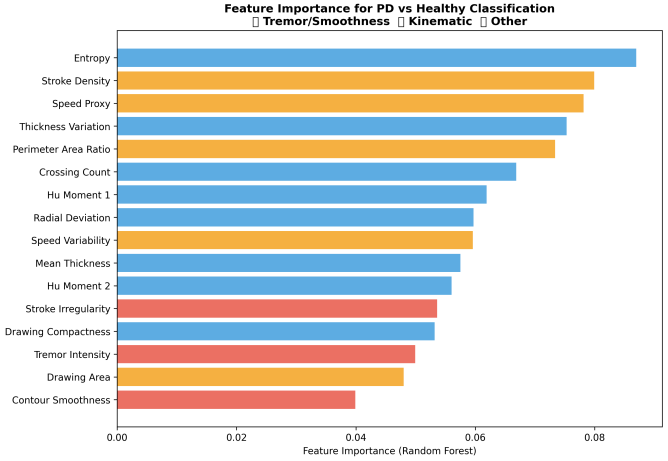


Fig. 3. Part A: Feature importance (Random Forest). Kinematic proxy features (orange) and structural features (blue) dominate.

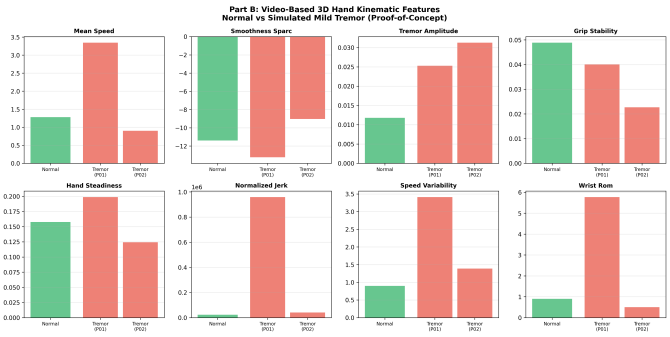


Fig. 4. Part B: 3D hand kinematic features comparing normal drawing (green) with simulated mild tremor (red). Large differences are consistent across participants.

independent samples t-tests revealed two statistically significant features (Table III).

SPARC smoothness was significantly more negative during micrographia ($p = 0.019$, Cohen's $d = 3.2$), indicating rougher movement. Drawing duration was significantly longer ($p = 0.031$, Cohen's $d = 3.14$). Importantly, these results demonstrate that micrographia can be detected from hand kinematics alone—without analyzing the written text—making the approach language-agnostic and independent of text visibility (Fig. 5).

4) Bradykinesia Observation: Under simulated bradykinesia, tremor amplitude increased substantially (0.083 vs. 0.035 baseline) and smoothness degraded (SPARC: -8.79 vs. -11.31 baseline). While formal statistical testing was not feasible at $n = 2$ per condition, the direction and magnitude are consistent with expected PD motor signatures.

V. Discussion

A. Complementary Nature of Product and Process Analysis

Product and process analysis capture different aspects of PD motor impairment. **Product analysis (Part A)** captures the cumulative effect on the final drawing (69.7% accuracy,

TABLE III
Part B: Micrographia Detection (Normal vs. Simulated)

Feature	Normal	Micro.	p	d
SPARC Smoothness	-11.31 ± 0.31	-12.35 ± 0.34	0.019*	3.20
Duration (s)	22.12 ± 6.53	38.96 ± 3.88	0.031*	3.14

$*p < 0.05$. Large effect sizes (Cohen's $d > 3$) noted despite small n. Results are proof-of-concept pending clinical validation.

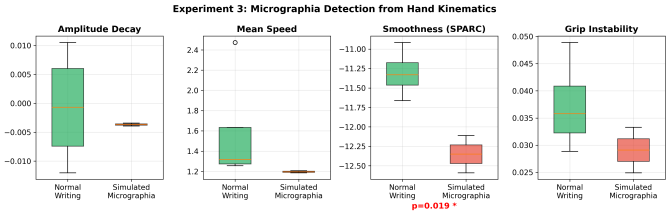


Fig. 5. Part B: SPARC smoothness significantly distinguishes normal writing from simulated micrographia ($p = 0.019$*), without requiring analysis of the written output.

validated on 204 real drawings), but cannot detect tremor that was motor-compensated, or hesitations that did not alter the final stroke. **Process analysis (Part B)** captures the moment-to-moment motor state during drawing—tremor, grip stability, smoothness, and velocity dynamics—detecting motor phenomena before they manifest in the product. Our proof-of-concept shows $> 1000\%$ increases in kinematic features under mild tremor, suggesting high sensitivity. The combined framework offers product analysis for validated large-scale screening and process analysis for daily monitoring at zero additional hardware cost.

B. Key Findings

Micrographia from hand kinematics. Detecting micrographia through SPARC smoothness ($p = 0.019$) without analyzing written text is clinically meaningful: it makes detection language-agnostic, robust to partial occlusion of the writing surface, and applicable regardless of writing content. The insight that *how* a task is executed carries as much diagnostic signal as *what* is produced [13] motivates extending this principle to other motor assessment tasks beyond drawing.

Feature sensitivity. The Part B effect sizes (e.g., $+2101\%$ jerk increase) suggest that 3D hand kinematics may ultimately exceed the sensitivity of 2D product analysis, where Cohen's d ranged from 0.31 to 0.51. The process modality captures 3D hand tremor and grip dynamics that the 2D output cannot represent.

Accessibility. Part B requires only a consumer smartphone—present in approximately 85% of the global population—making daily monitoring technically feasible in resource-limited settings where specialized tablets are unavailable.

C. Broader Implications

Our framing of PD motor assessment as a *sketch process analysis* problem opens a productive connection to HCI and

sketch recognition research. The kinematic features we extract—smoothness, velocity profiles, temporal dynamics, and spatial amplitude—are structurally identical to the features used to analyze drawing fluency and motor expertise in freehand input systems [8], [13]. Process analysis in Part B is equivalent to studying *how* someone draws rather than *what* they drew, a distinction central to sketch recognition systems [10], [11]. This analogy suggests that feature engineering advances from the sketch recognition literature can be systematically applied to clinical motor assessment—a cross-disciplinary direction that warrants dedicated future investigation.

D. Limitations

- **Part A accuracy**: 66.7–69.7% is moderate. Deep learning approaches may improve performance but were outside the scope of this feasibility study.
- **Part B sample size**: With $n = 2$ healthy participants, Part B is strictly proof-of-concept. Statistical significance was achieved for micrographia (large effect sizes compensate), but clinical validation with diagnosed PD patients is essential.
- **Simulated vs. real PD**: Physical symptom simulation in healthy participants does not fully replicate the neurological mechanisms of PD.
- **Detection variability**: Hand tracking ranged from 10.7% to 95.6% depending on lighting and motion blur. Strong tremor conditions had lower detection rates due to blur, requiring future optimization.

E. Future Work

Immediate priorities: (1) clinical validation with diagnosed PD patients (Hoehn & Yahr I–IV); (2) larger samples ($n > 30$ per condition); (3) longitudinal daily monitoring studies; (4) deep temporal models including state-space architectures [22] for hand trajectory classification; (5) unified product-plus-process feature fusion; and (6) a real-time smartphone application providing clinician-interpretable motor monitoring output.

VI. Conclusion

We presented a two-part feasibility study of complementary approaches to Parkinson's motor assessment through drawing analysis. Part A validated that 16 image-based sketch features distinguish PD from healthy controls (best: 69.7% SVM accuracy, $n = 204$), with image entropy and stroke density as top discriminators. Part B demonstrated that novel 3D video-based hand kinematics—requiring only a consumer smartphone—detect simulated PD motor symptoms with large magnitude differences ($> 1000\%$ for tremor features) and achieve statistically significant micrographia detection from hand movement alone ($p = 0.019$, Cohen's $d = 3.2$).

The transition from drawing product to drawing process analysis represents a paradigm shift: from analyzing *what* patients drew (requiring specialized hardware) to analyzing *how* they draw (requiring only a phone camera). Future clinical validation with diagnosed PD patients will determine whether these proof-of-concept results generalize to real-world motor assessment and longitudinal monitoring.

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